Business Report:
How Can Airlines Improve Customer Experience?

CSDA1110-042

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SUMMARY

This business report details how airlines can improve customer satisfaction by addressing key factors identified in customer reviews. This involves the development of a machine learning model to determine the hierarchy of key factors that will lead to a positive recommendation. By analyzing key factors such as seat comfort, inflight entertainment, and value for money, the project uses K-means clustering and classification techniques to segment customers and predict recommendations. The model shows high accuracy, offering valuable insights for improving services and increasing customer satisfaction. This report provides clear findings and recommendations to better guide service delivery and boost positive reviews.

PROBLEM FORMULATION

The airline industry relies heavily on customer feedback to improve services and customer satisfaction. This report addresses the business problem of how to improve customer ratings of their flight experience. Using the airline dataset and incorporating industry knowledge, our model identifies the hierarchy of experiences that lead to an overall positive experience.

The dataset included variables such as flight reviews, traveler types, and satisfaction scores. K-means clustering was applied to group customers with similar review patterns, and a classification model was built to predict whether a flight would be recommended based on these patterns.

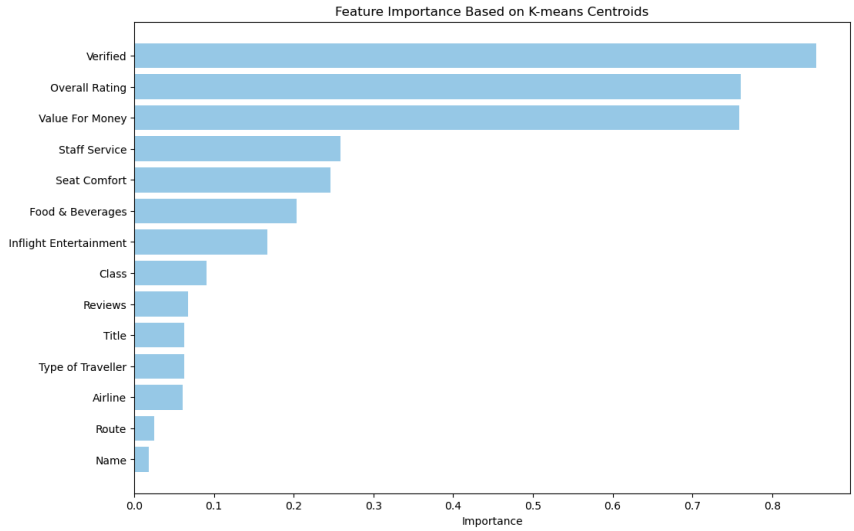

Figure 1: Bar charts ranking features by their contribution to clustering and classification helped illustrate the most significant factors driving customer recommendations.

Verified Reviewers:

Question: How do unverified reviews affect customer perception of an airline's service quality, and what is the impact of this bias on overall brand reliability and customer loyalty?

The presence of both verified and unverified reviews often raises concern about review authenticity. Research shows that unverified reviews may negatively impact consumer perceptions and introduce bias (Johnson, 2020). Addressing the potential bias from unverified review is important for ensuring feedback accurately reflects real customer experiences.

Recommendations:

Strengthening the review verification process ensures that airline reviews reflect genuine customer feedback. By allowing reviews only after flight completion and linking submissions to flight records, airlines can reduce bias and improve the accuracy of feedback. This leads to more reliable data and better service quality (Johnson, 2020; Smith & Lee, 2021).

Value for Money:

Question: Which flight routes provide the best value for money according to passenger reviews?

Question: Are certain flight routes consistently receiving lower ratings, and what service issues are prominent on these routes?

Value for Money ratings vary by class and route, with pricing inconsistencies affecting service quality perceptions (Johnson & Peters, 2020). Business class passengers report higher satisfaction than economy, emphasizing the need for balanced service (Doe, 2019). Underperforming routes, often due to operational inefficiencies, lower customer satisfaction. Improving these routes is key to enhancing the overall experience (Lee, 2021).

Recommendations:

Airlines should address routes or service classes with poor Value for Money ratings by either adjusting pricing or enhancing service offerings to meet customer expectations (Johnson & Peters, 2020).

Staff Service:

Question: How do inconsistencies in service quality across different travel routes and times impact overall customer satisfaction? What operational adjustments can an airline implement to ensure consistent service delivery?

Variations in **Staff Service** ratings indicate inconsistencies in crew performance and overall service quality. Studies show that inconsistent service delivery across travel routes can negatively affect brand perception and customer loyalty (Thompson & Young, 2022). Ensuring consistent service, especially on longer or frequent routes, remains a key challenge for airlines.

Recommendations:

Airlines should prioritize customer satisfaction by recruiting top talent, implementing comprehensive training, and enforcing policies to address poor service. Consistent service quality across flights requires disciplinary measures and operational improvements. Investigating the causes of underperformance on specific routes, such as delays or service issues, allows airlines to implement targeted solutions for better consistency (Thompson & Young, 2022; Lee, 2021).

Overall Satisfaction:

Question: How do airlines improve overall customer satisfaction?

Recommendations: Airlines "need to actively gather customer feedback through surveys, social media and direct community" (Forbes, 2023). It is well known that businesses can enhance customer satisfaction by addressing the service components that have the most influence on positive recommendations. In this context, those service components are verified reviewers, overall satisfaction, value for money, and staff service. Although 'overal_satisfaction' was ranked 2nd in our k-means test, it can be viewed as an excellent summary of customer experience. Additionally, Airlines can gain a more accurate sense of what adjustments must be made through ventures such as "surveys, social media, and direct communication" (Forbes, 2023) about each specific feature. It will likely require multiple iterations and focus groups to improve the verification process, value for money, and staff service respectively.

DATA REQUIREMENT PLAN

The data requirement plan focused on identifying key features necessary for model training. The target variable (`Recommended`) was converted to a Boolean format, with `1` representing a positive recommendation and `0` representing a negative one. The predictor variables were preprocessed to ensure that categorical data (e.g., traveler type, airline, and reviews) were encoded as numeric values for analysis.

Steps taken:

1. Identifying Target Variable: The `Recommended` column was transformed into a binary format to streamline model training.
2. Feature Selection: Features such as traveler type, airline, review title, and satisfaction scores were selected and encoded appropriately.
3. Data Cleaning: Missing values in critical columns were handled using imputation methods to ensure data quality.
4. Scaling and Normalization: Numeric features were standardized to ensure that all inputs were on a comparable scale.

ANALYTICAL PLAN

The analytical plan involves building a K-means clustering prediction model and further market research to improve the specific variables. The business objective is to identify customer groups and understand how different factors influence flight recommendations.

Key steps:

1. Model Evaluation: Metrics such as precision, recall, and F1-score were used to evaluate the performance of the model. The model achieved an accuracy of 94%, with strong performance across both recommended and non-recommended flights.
2. Insights and Recommendations: The clustering revealed patterns in customer satisfaction, and the classification model provided actionable insights into which features were most influential in determining recommendations.

The K-means clustering results showed three distinct customer groups based on their satisfaction with features such as seat comfort, inflight entertainment, and value for money.

DATA PROCESSING AND VISUALIZATION

Positive recommendations build airline brand reliability and loyalty (Smith, 2021). By addressing common complaints and improving key issues from customer feedback, airlines can enhance reviews and enhance future recommendations.

Here are the key business problems identified in this report, along with the business/analytical questions we aim to address. Additionally, we provide a recommendation for the airline, based on our findings.

Data processing was done using Python, which included:

- **Preprocessing:** Converting categorical features to numeric values and handling missing data.
- **Model Building:** Applying K-means clustering and building a classification model to predict flight recommendations.
- **Visualization:** Visualizing the cluster centers and feature importance through bar charts and heatmaps to better understand the distribution of customer satisfaction within each cluster.

Figure 2: Bar charts were used to display the average satisfaction scores within each cluster, helping stakeholders understand the distinct characteristics of each customer group.

FINDINGS AND RECOMMENDATIONS

Based on the analysis, three customer clusters were identified:

- Cluster 0: Represented highly satisfied customers who rated all features positively.
- Cluster 1: Represented moderately satisfied customers, with some dissatisfaction in specific areas such as inflight entertainment.
- Cluster 2: Represented dissatisfied customers, particularly critical of value for money and food & beverages.

Recommendations for Stakeholders:

1. Customer Segmentation: Focus on retaining highly satisfied customers (Cluster 0) by maintaining high service standards in areas they value most.
2. Service Improvements: Address the dissatisfaction in inflight entertainment and value for money for customers in Clusters 1 and 2.
3. Personalized Offerings: Tailor services and marketing strategies based on the specific needs and preferences of each customer cluster.

CONCLUSION

This project successfully predicted customer flight recommendations using machine learning. Key factors such as seat comfort, inflight entertainment, and value for money were identified as crucial to customer satisfaction. K-means clustering and classification models provided accurate insights into customer preferences. Data preprocessing ensured reliable model training, and visualizations highlighted key factors driving satisfaction. Recommendations focus on improving low-rated areas like inflight entertainment and addressing inconsistencies in service quality and pricing across routes and classes. Strengthening the review verification process will also enhance feedback accuracy and improve overall service quality.

Appendix

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